National Technical University of Ukraine
“Igor Sikorsky Kyiv Polytechnic Institute”
Faculty of Informatics and Computer Science
Department of Information Systems and Technologies

Lab work No4

Data visualization with matplotlib and Seaborn

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Tasks

Create a Python program that does task according to your number in the group list.

Make a report in doc-file. Report must contain:
— Title;

— Program code;
— Results (screenshots) of code execution.

Demonstrate the working program and answer questions about theoretical information and program working.

File merc.csv.

1. Construct bar charts to display a) the number of mercedes of each model; b) the average car price of each model; c) the average car price of each model for each transmission type.

2. Build a price histograms, overall and for each transmission type
.
3. Build a boxplot of the fuel consumption (mpg, miles per gallon), overall and for each fuel type, determine whether outliers are present.

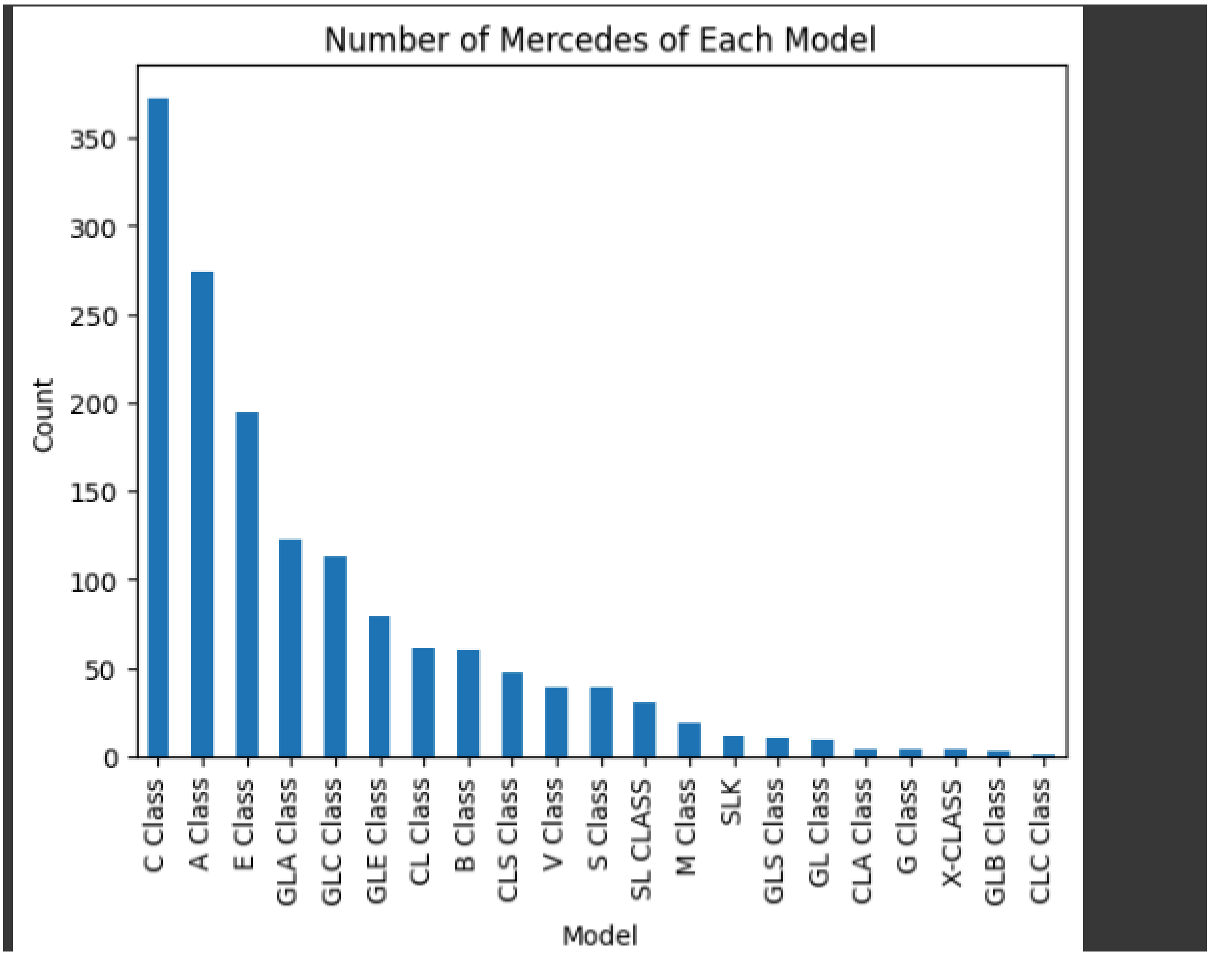
4. Using scatter plots determine whether there is relationship between a) price and mileage; b) mpg and engine size.

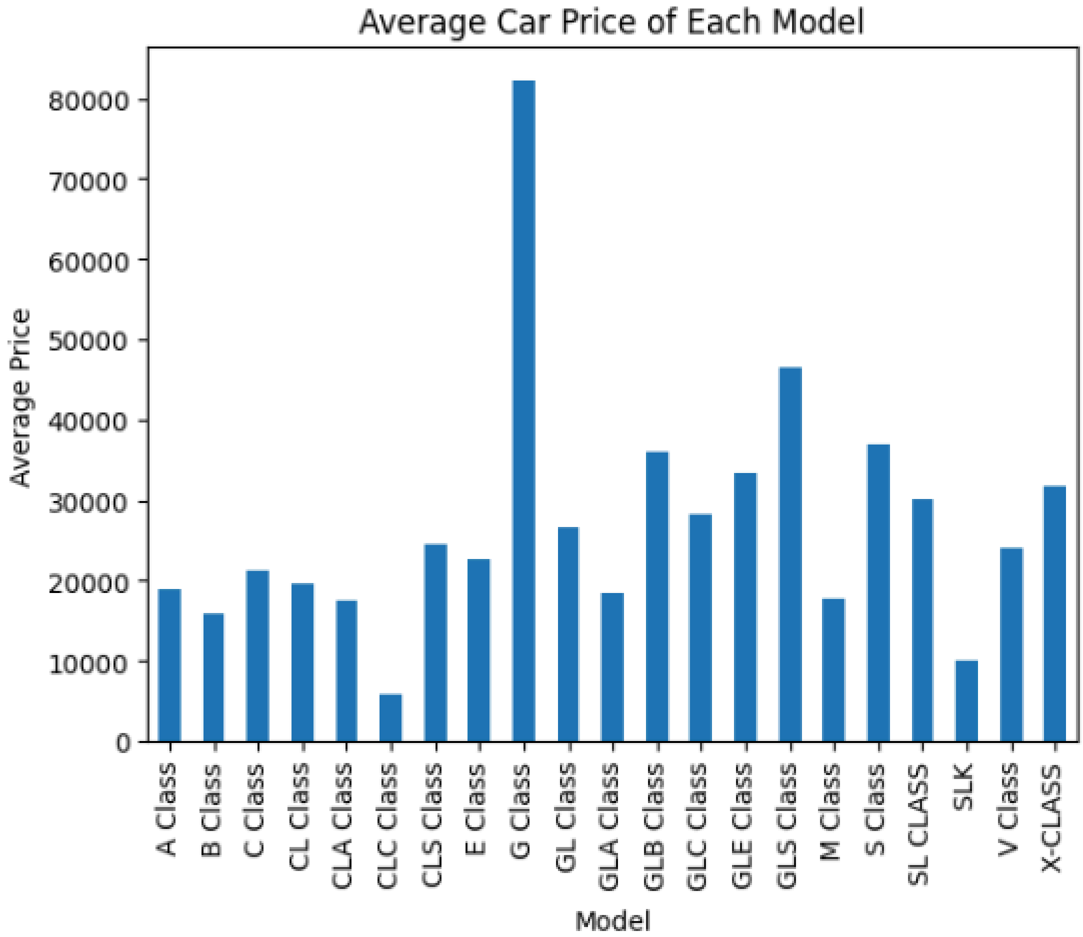
My Full code:

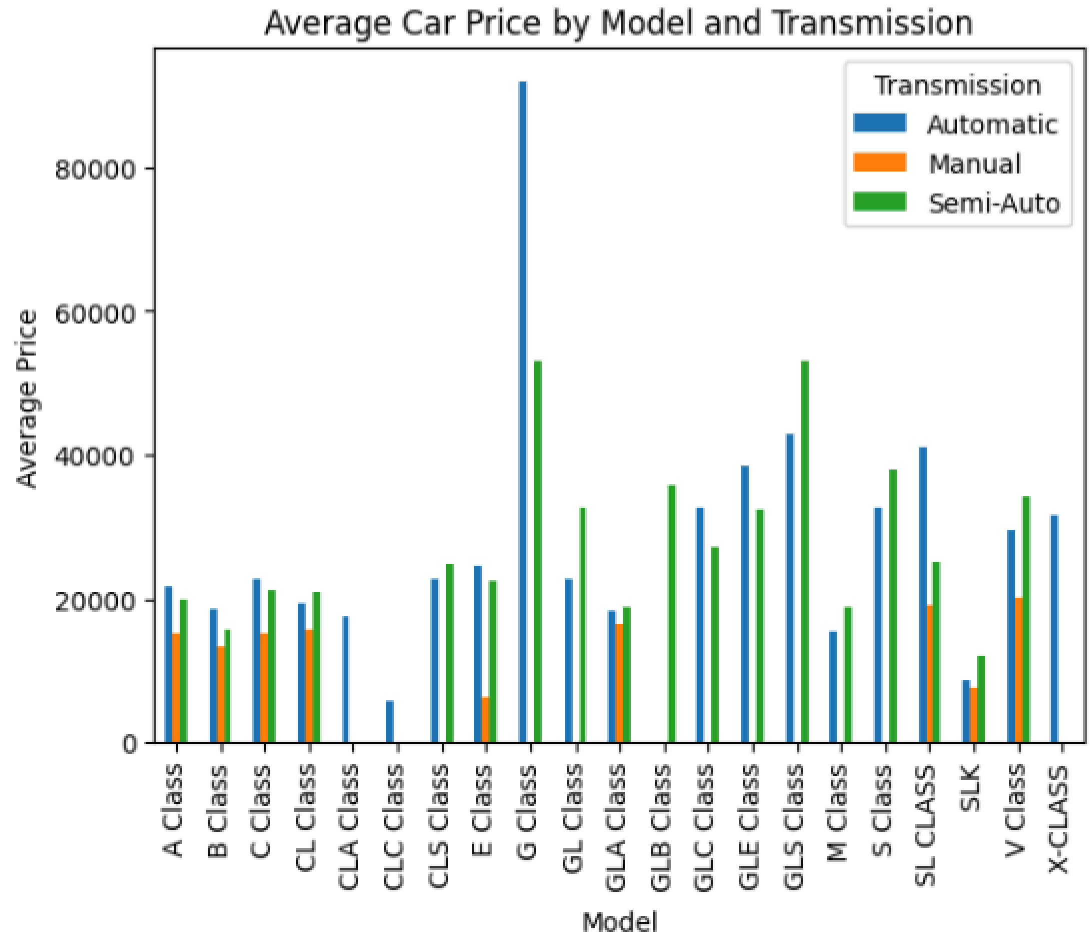
```python
import pandas as pd
import matplotlib.pyplot as plt

# Read the data from the CSV file
df = pd.read_csv('merc.csv')

# 1a. Bar chart - Number of Mercedes of each model
model_counts = df['model'].value_counts()
model_counts.plot(kind='bar', title='Number of Mercedes of Each Model')
plt.xlabel('Model')
plt.ylabel('Count')
plt.show()

# 1b. Bar chart - Average car price of each model
avg_price_by_model = df.groupby('model')['price'].mean()
avg_price_by_model.plot(kind='bar', title='Average Car Price of Each Model')
plt.xlabel('Model')
plt.ylabel('Average Price')
plt.show()

# 1c. Bar chart - Average car price of each model for each transmission type
avg_price_by_transmission = df.groupby(['model', 'transmission'])['price'].mean().unstack()
avg_price_by_transmission.plot(kind='bar', title='Average Car Price by Model and Transmission')
plt.xlabel('Model')
plt.ylabel('Average Price')
plt.legend(title='Transmission')
plt.show()
```

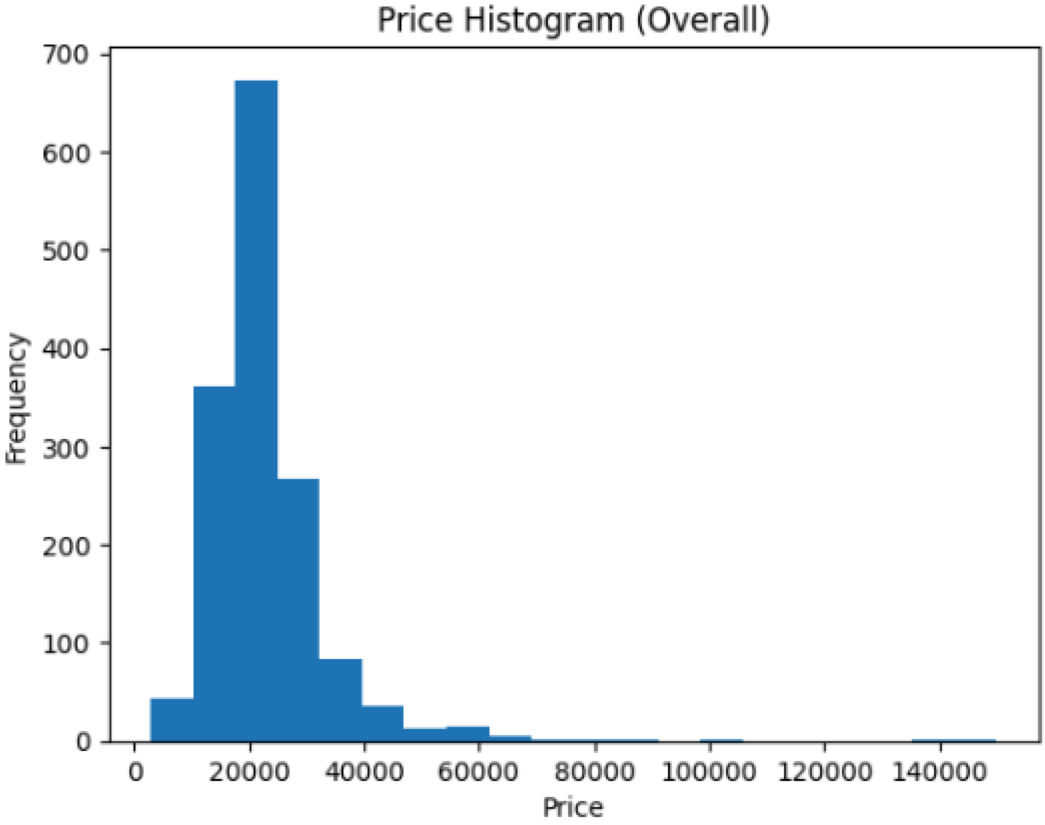
```python
# 2. Price histograms overall and for each transmission type
df['price'].plot(kind='hist', bins=20, title='Price Histogram (Overall)')
plt.xlabel('Price')
plt.ylabel('Frequency')
plt.show()

for transmission_type in df['transmission'].unique():
    df[df['transmission'] == transmission_type]['price'].plot(kind='hist', bins=20,
                                                              title=f'Price Histogram ({transmission_type} Transmission)')

    plt.xlabel('Price')
    plt.ylabel('Frequency')
    plt.show()

# 3. Boxplot of fuel consumption (mpg)
df.boxplot(column='mpg', by='fuelType', grid=False)
plt.title('Boxplot of Fuel Consumption (mpg) by Fuel Type')
plt.suptitle('')
plt.xlabel('Fuel Type')
plt.ylabel('MPG')
plt.show()

# 4a. Scatter plot - Price vs. Mileage
plt.scatter(df['mileage'], df['price'])
plt.title('Scatter Plot: Price vs. Mileage')
plt.xlabel('Mileage')
plt.ylabel('Price')
plt.show()
```

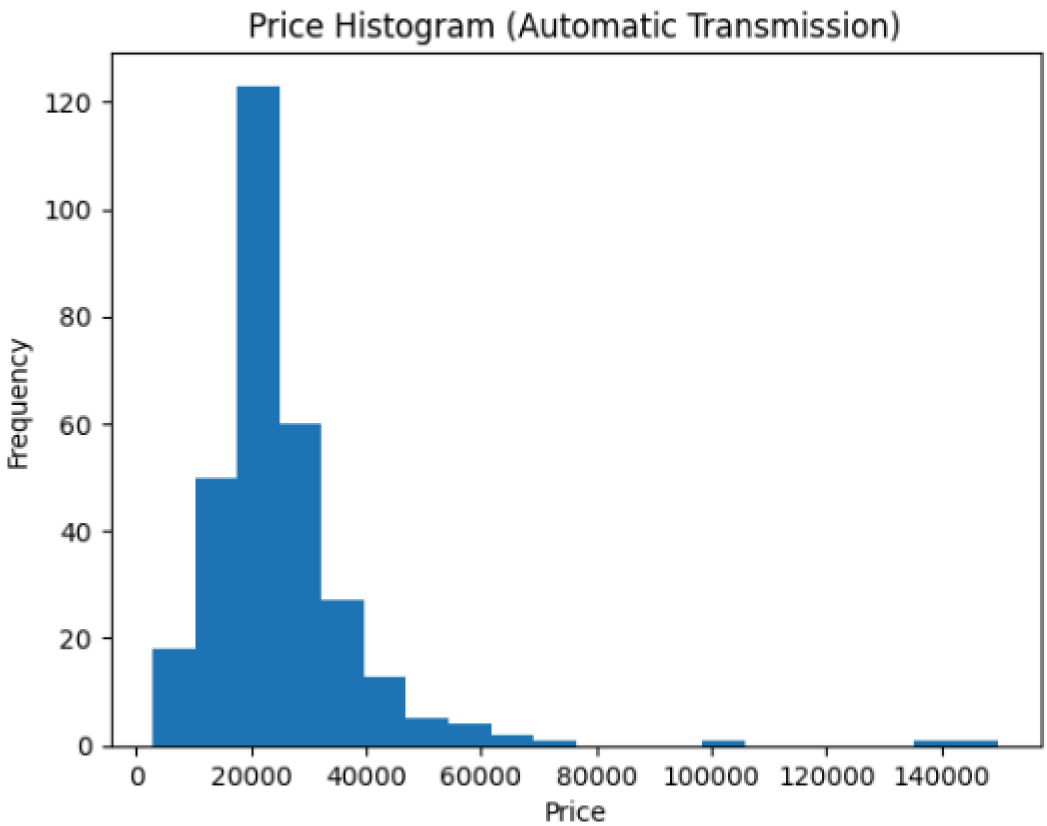
```python
# 4b. Scatter plot - MPG vs. Engine Size
plt.scatter(df['mpg'], df['engineSize'])
plt.title('Scatter Plot: MPG vs. Engine Size')
plt.xlabel('MPG')
plt.ylabel('Engine Size')
plt.show()
```

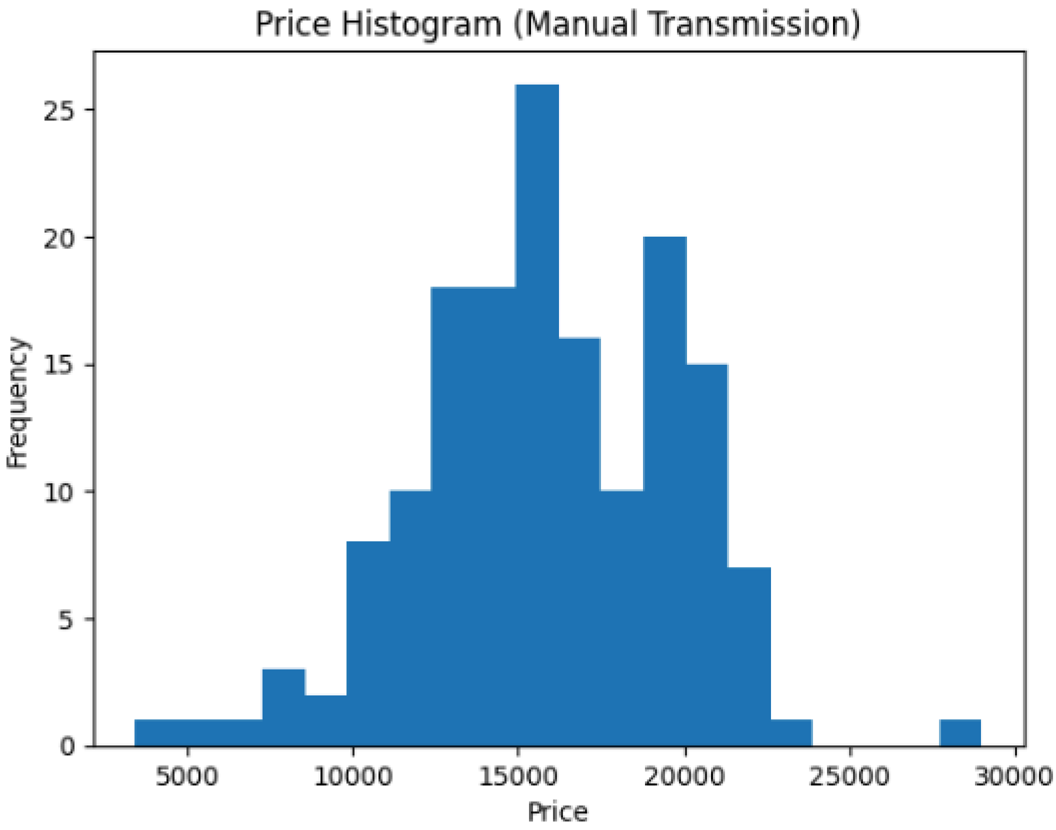

Number of Mercedes of Each Model
Count
Model
0
50
100
150
200
250
300
350
C Class
A Class
E Class
GLA Class
GLC Class
GLE Class
CL Class
B Class
CLS Class
V Class
S Class
SL CLASS
M Class
SLK
GLS Class
GL Class
CLA Class
G Class
X-CLASS
GLB Class
CLC Class

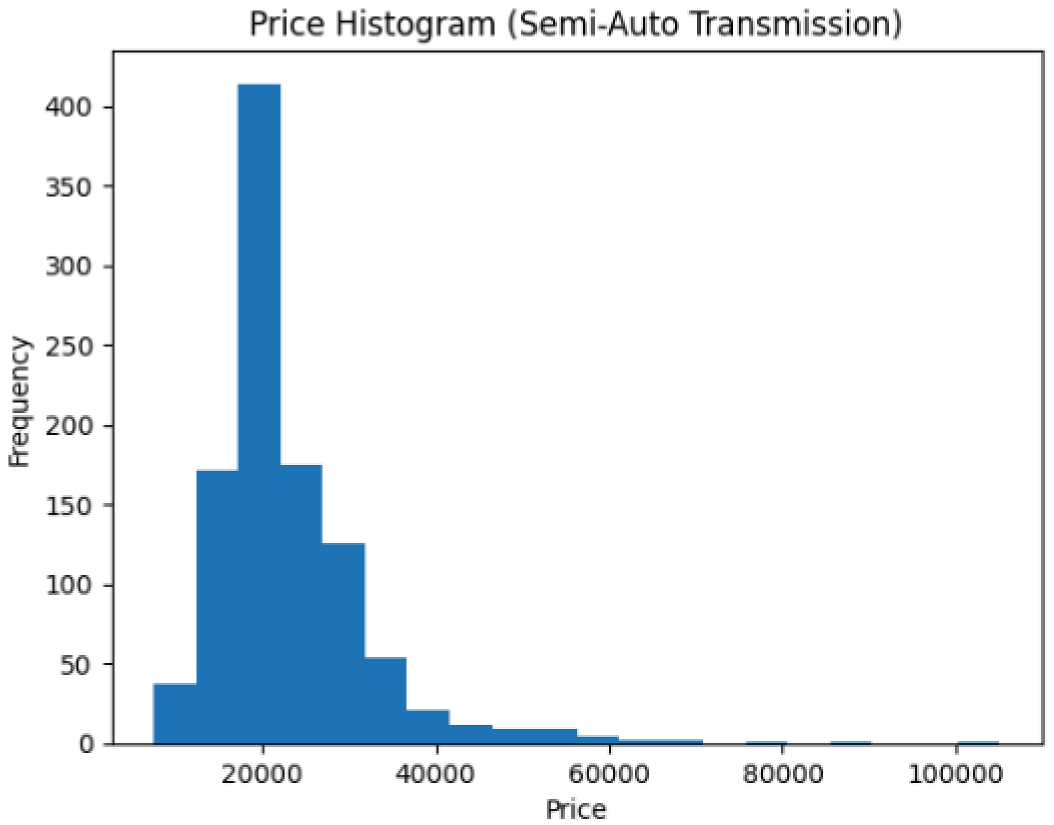

Average Car Price of Each Model
Average Price
0
10000
20000
30000
40000
50000
60000
70000
80000
A Class
B Class
C Class
CL Class
CLA Class
CLC Class
CLS Class
E Class
G Class
GL Class
GLA Class
GLB Class
GLC Class
GLE Class
GLS Class
M Class
S Class
SL CLASS
SLK
V Class
X-CLASS
Model

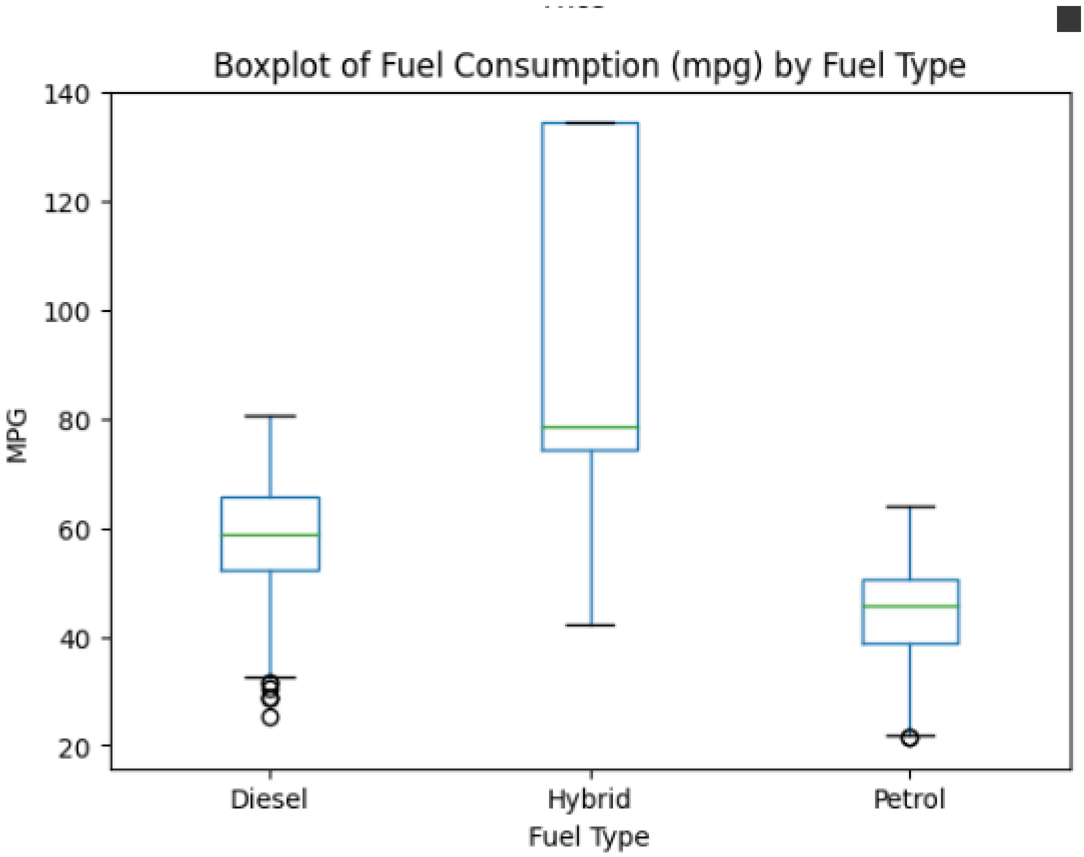

Average Car Price by Model and Transmission
Transmission
Automatic
Manual
Semi-Auto
Average Price
0
20000
40000
60000
80000
A Class
B Class
C Class
CL Class
CLA Class
CLC Class
CLS Class
E Class
G Class
GL Class
GLA Class
GLB Class
GLC Class
GLE Class
GLS Class
M Class
S Class
SL CLASS
SLK
V Class
X-CLASS
Model

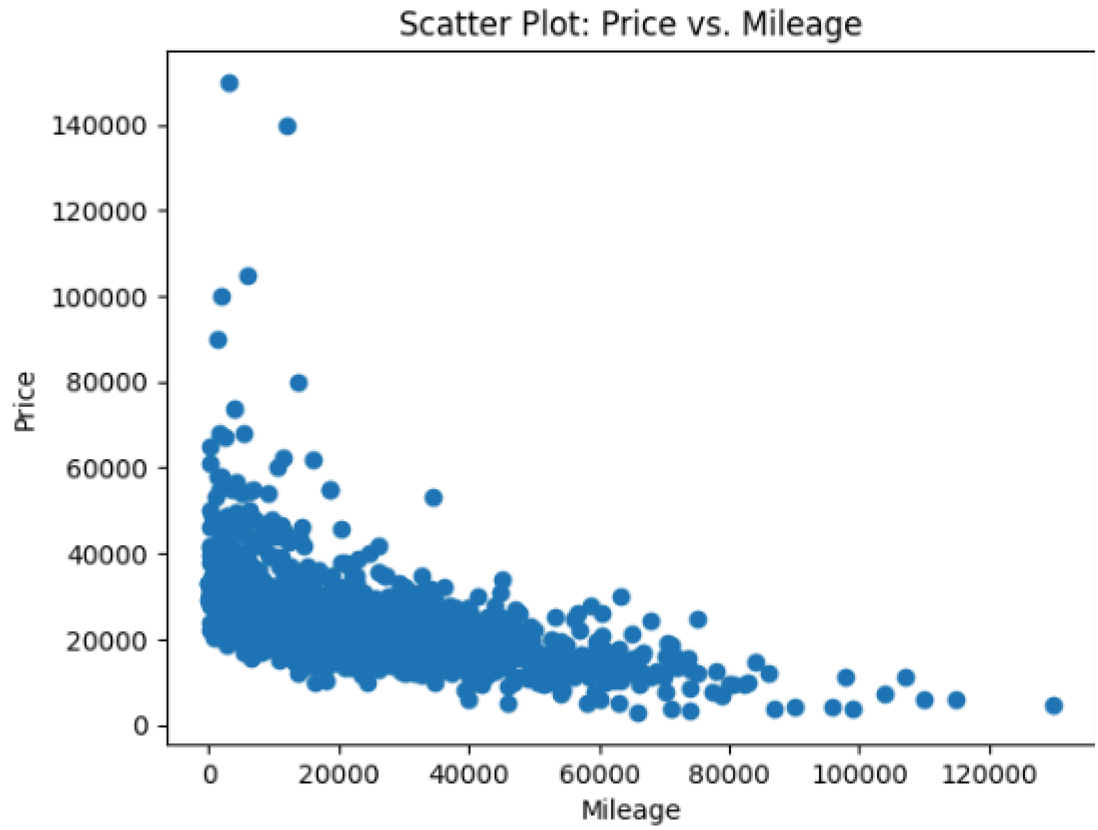

Price Histogram (Overall)
Frequency
Price
0
100
200
300
400
500
600
700
0
20000
40000
60000
80000
100000
120000
140000

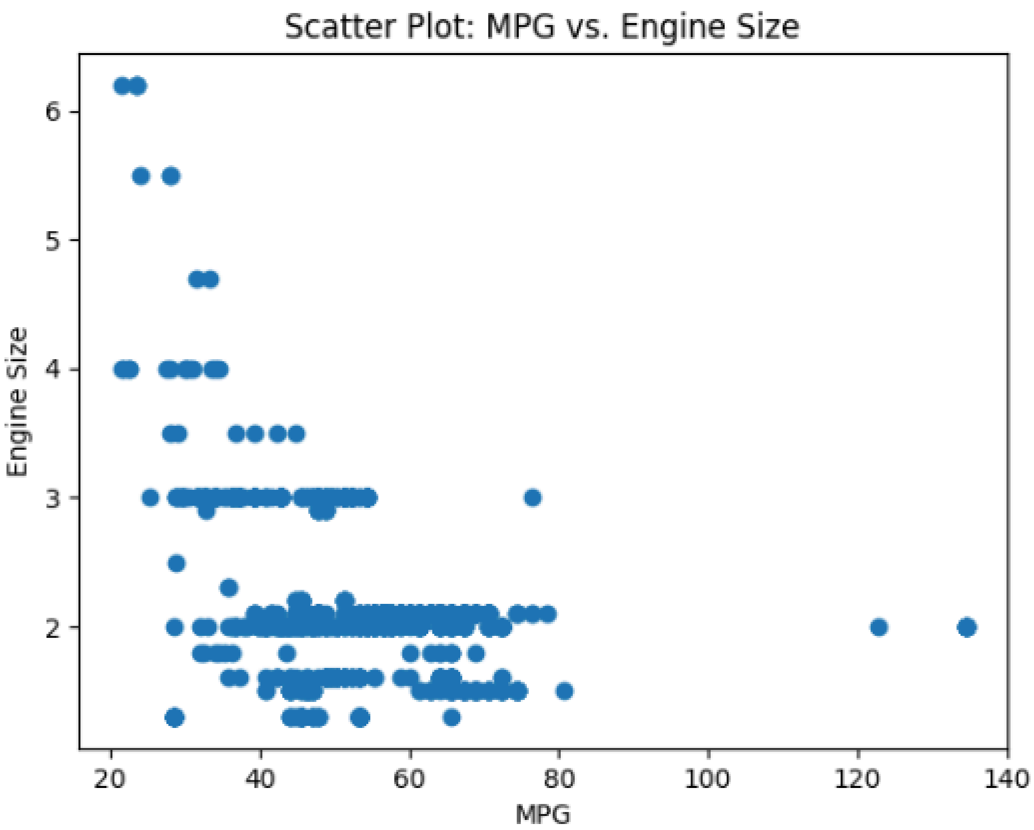

Price Histogram (Automatic Transmission)
Frequency
Price
0
20
40
60
80
100
120
0
20000
40000
60000
80000
100000
120000
140000


Price Histogram (Manual Transmission)
Frequency
0
5
10
15
20
25
5000
10000
15000
20000
25000
30000
Price


Price Histogram (Semi-Auto Transmission)
Frequency
0
50
100
150
200
250
300
350
400
20000
40000
60000
80000
100000
Price


Boxplot of Fuel Consumption (mpg) by Fuel Type
140
120
100
80
60
40
20
MPG
Diesel
Hybrid
Petrol
Fuel Type


Scatter Plot: Price vs. Mileage
Price
Mileage
0
20000
40000
60000
80000
100000
120000
140000


Scatter Plot: MPG vs. Engine Size
Engine Size
MPG
2
3
4
5
6
20
40
60
80
100
120
140